

Divided Differences and Newton's Divided Difference Interpolation

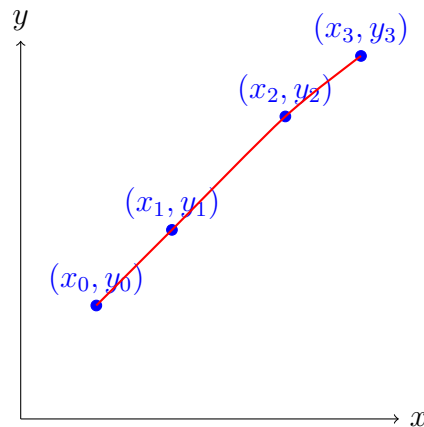
Teacher-Style Notes

1 Introduction

Interpolation is the process of estimating values of a function at points within the range of known data points. It allows us to approximate a continuous curve when only discrete points are known.

Divided differences provide a systematic way to construct interpolating polynomials incrementally, and are particularly useful for Newton's interpolation method.

Diagram of Data Points



2 Divided Differences

2.1 Definition

Given $n + 1$ data points $(x_0, y_0), \dots, (x_n, y_n)$:

- **Zero-order:** $f[x_i] = y_i$
- **First-order:**

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$$

- **Higher-order:** recursively,

$$f[x_i, x_{i+1}, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

Intuition: Each higher-order divided difference measures the change of the previous order differences, capturing slope, curvature, and higher-level variations.

3 Newton's Divided Difference Formula

The Newton interpolating polynomial is:

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots + f[x_0, \dots, x_n](x - x_0) \dots (x - x_{n-1})$$

Explanation of terms:

- $f[x_0]$ = value at first point
 - $f[x_0, x_1](x - x_0)$ = linear term
 - $f[x_0, x_1, x_2](x - x_0)(x - x_1)$ = quadratic term
 - Higher-order terms follow similarly
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4 Simplified Notation

The polynomial can be compactly written as:

$$P_n(x) = \sum_{k=0}^n f[x_0, x_1, \dots, x_k] \prod_{j=0}^{k-1} (x - x_j)$$

5 Error Analysis

The interpolation error at a point x is:

$$R_n(x) = f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i), \quad \xi \in [\min(x_i), \max(x_i)]$$

Intuition: The error depends on the next derivative of f and the product of differences from all known points.

6 Worked Example

Data Points

$$(x_0, y_0) = (1, 2), \quad (x_1, y_1) = (2, 3), \quad (x_2, y_2) = (4, 7)$$

Step 1: Compute Divided Differences

x_i	$f[x_i]$	1st	2nd
1	2	$f[1, 2] = (3 - 2)/(2 - 1) = 1$	$f[1, 2, 4] = (2 - 1)/(4 - 1) = 1/3$
2	3	$f[2, 4] = (7 - 3)/(4 - 2) = 2$	
4	7		

Step 2: Construct Newton Polynomial

$$\begin{aligned}
 P_2(x) &= f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\
 &= 2 + 1(x - 1) + \frac{1}{3}(x - 1)(x - 2) \\
 &= \frac{1}{3}x^2 - x + 3
 \end{aligned}$$

Verification:

$$P_2(1) = 2, \quad P_2(2) = 3, \quad P_2(4) = 7$$

7 Summary Table

Concept	Formula / Notes
Zero-order divided difference	$f[x_i] = y_i$
First-order divided difference	$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$
Higher-order divided differences	$f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$
Newton interpolating polynomial	$P_n(x) = \sum_{k=0}^n f[x_0, \dots, x_k] \prod_{j=0}^{k-1} (x - x_j)$
Error formula	$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$
Advantages	Incremental, flexible spacing, easy evaluation
Limitations	Oscillations for high-degree, sensitive to noise, overfitting

Newton's Forward Difference Interpolation – Notes

1. Introduction

Newton's Forward Difference Method is used when:

- The data points are **equally spaced**, i.e. $x_{i+1} - x_i = h$.
- The value to be estimated lies **near the beginning** of the table.

It rewrites the interpolation polynomial using forward differences:

$$\Delta f(x_0), \Delta^2 f(x_0), \Delta^3 f(x_0), \dots$$

Let

$$p = \frac{x - x_0}{h}.$$

2. Forward Difference Table

Forward differences are defined as:

$$\begin{aligned}\Delta f(x_i) &= f(x_{i+1}) - f(x_i), \\ \Delta^2 f(x_i) &= \Delta f(x_{i+1}) - \Delta f(x_i), \\ \Delta^k f(x_i) &= \Delta^{k-1} f(x_{i+1}) - \Delta^{k-1} f(x_i).\end{aligned}$$

3. Newton's Forward Difference Formula

$$P_n(x) = f(x_0) + p \Delta f(x_0) + \frac{p(p-1)}{2!} \Delta^2 f(x_0) + \frac{p(p-1)(p-2)}{3!} \Delta^3 f(x_0) + \dots + \frac{p(p-1) \cdots (p-n+1)}{n!} \Delta^n f(x_0)$$

4. When to Use Forward vs General Newton Form

- Use **Newton Forward** if:
 - x is near the beginning,
 - points are equally spaced.
- Use **Newton's General Divided Difference** if:
 - points are not equally spaced, or
 - x is not close to the first point.

5. Forward Difference Error Formula

Assume f is smooth and $(n+1)$ times differentiable.

$$R_n(x) = \frac{h^{n+1}}{(n+1)!} p(p-1)(p-2) \cdots (p-n) f^{(n+1)}(\xi),$$

for some ξ in the interval containing the data points.

6. Error Bound (Practical Use)

If an upper bound on $|f^{(n+1)}(x)|$ is known:

$$|R_n(x)| \leq \frac{h^{n+1}}{(n+1)!} |p(p-1)(p-2) \cdots (p-n)| \max_{\xi} |f^{(n+1)}(\xi)|.$$

In real numerical problems, $f^{(n+1)}$ is usually unavailable, so the bound is used for:

- theoretical justification,
- comparing which interpolation degree is sufficient,
- estimating how errors grow for large n .

7. Summary

- Forward method: best for equally spaced data near x_0 .
- Uses finite differences instead of divided differences.
- Parameter $p = \frac{x - x_0}{h}$ is computed once for each x .
- Error term resembles Taylor remainder with forward differences.

Newton's Backward Difference Interpolation

1. When to Use

- Target value x is near the last data point x_n .
- Equally spaced data points.

2. Normalized Variable

$$p = \frac{x - x_n}{h}, \quad h = x_{i+1} - x_i$$

3. Backward Difference Table

Backward differences (∇) are computed from the last row upward:

$$\nabla f_n = f_n - f_{n-1}, \quad \nabla^2 f_n = \nabla f_n - \nabla f_{n-1}, \quad \dots$$

x	$f(x)$	∇f	$\nabla^2 f$	$\nabla^3 f$
x_0	f_0			
x_1	f_1			
x_2	f_2	∇f_3		
x_3	f_3	∇f_3	$\nabla^2 f_3$	$\nabla^3 f_3$

4. Polynomial Formula

$$P_n(x) = f_n + p\nabla f_n + \frac{p(p+1)}{2!}\nabla^2 f_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 f_n + \dots + \frac{p(p+1)\dots(p+n-1)}{n!}\nabla^n f_n$$

5. Error Term

$$R_n(x) = \frac{h^{n+1}}{(n+1)!} p(p+1)\dots(p+n) f^{(n+1)}(\xi), \quad \xi \in [x_0, x_n]$$

6. Key Points

- Backward differences are essentially the "mirror" of forward differences.
- Use table from forward method by re-indexing from the last row if allowed.
- Best when x is close to the last data point for better accuracy.
- Higher degree polynomial gives more accurate interpolation, but more computation.